

1. Method

Task1 - Spatial Domain

(1) Convolution

I implemented sliding window convolution using a three-layer loop, and it supports independent RGB three channels.

```
for i in tqdm(range(h), desc="Convolution"):
    for j in range(w):
        for c in range(3):
            region = padded[i:i+kh, j:j+kw, c]
            output[i, j, c] = np.sum(region * kernel)
```

(2) Laplacian Kernel

$$K_4 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

4-neighbor

$$K_8 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

8-neighbor

(3) Padding

I have implemented both edge padding methods:

- Zero-padding adds 0s around the edges (Main Result)
- Replicate-padding copies the edge pixels outwards.
- The padding size is 1, ensuring the output image is the same size as the original image.

(4) Sharpening formula

$$\text{sharpened} = \text{original} + \alpha * \text{laplacian_edge}$$

where α is an adjustable sharpening intensity parameter (here $\alpha=1$)

Task2 - Frequency Domain

(1) Fourier Transform

- Use `np.fft.fft2` to convert the image to the frequency domain.
- Use `np.fft.fftshift` to shift low-frequency components to the center of the spectrum, facilitating filter design and visualization.
- To observe spectral changes, calculate and store the Fourier Spectrum before and after filtering :

```
for c in range(3):
    spectrum_before[:, :, c] = np.log(1 + np.abs(f[:, :, c]))
```

(2) Frequency domain Laplacian filter

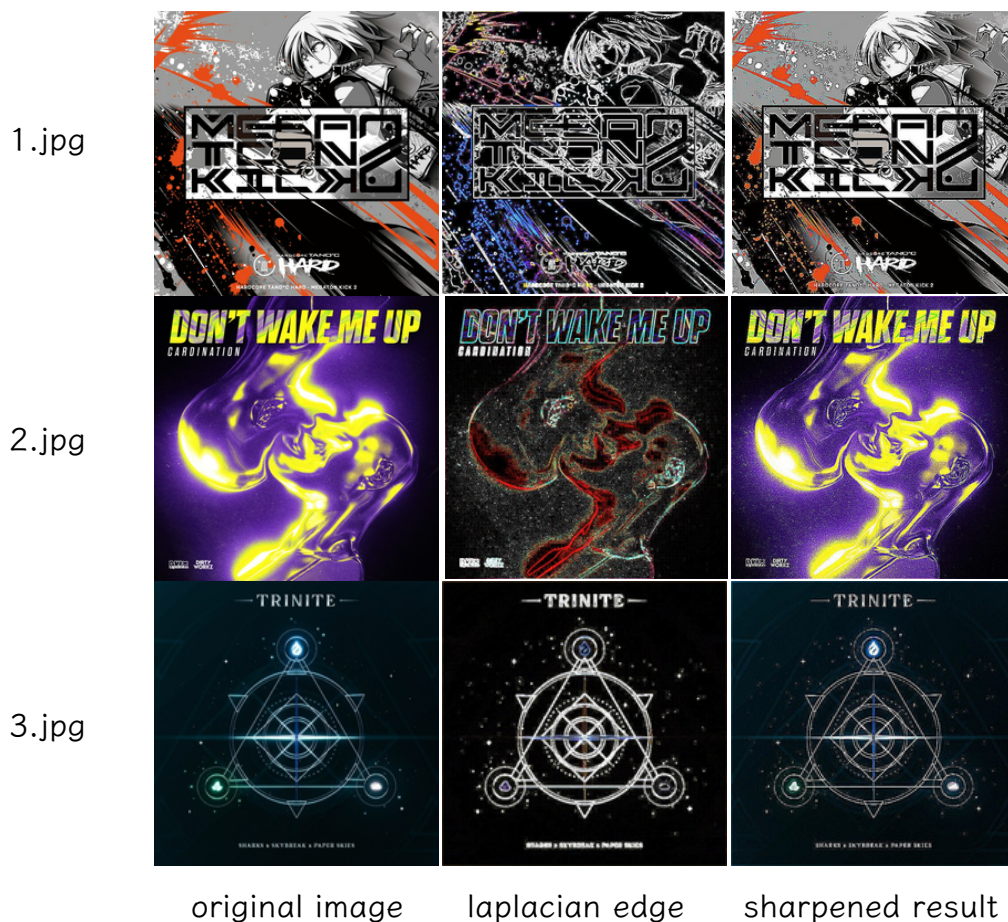
- Establish a frequency coordinate grid of the same size as the image:
- Laplacian frequency response: $H(u, v) = -(u^2 + v^2)$
- Normalize H to $[-1, 0]$: $H = H / \max(|H|)$

(3) Sharpening filter

$$H_{\text{sharpen}}(u, v) = 1 + \alpha * H(u, v)$$

(4) Inverse Fourier Transform

- Use `np.fft.ifftshift` to shift low frequencies back to the corner.
- Use `np.fft.ifft2` for inverse Fourier transform.
- Use `np.real` to extract the real part; the imaginary part represents the calculation error.

2. Result(1) Spatial Domain(2) Laplacian Kernel {4,8}

neighbor	4	8
Response direction		
Edge continuity	less	complete
Noise sensitivity	low	high

The sharpening effect of 8-neighbors is more significant.

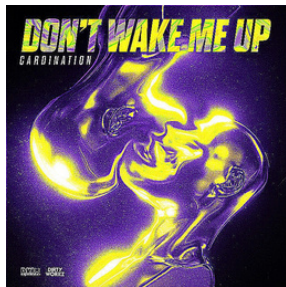
(3) Frequency Domain

alpha = 20.0



original image laplacian freq sharpened freq spectrum before spectrum after

In the filtered Fourier spectrum, a **black ring** can be observed around the central bright spot; the reason for this will be discussed later.

(4) Spatial v.s. Frequency

spatial



frequency

Domain	Spatial	Frequency
Sharpening effect	$\alpha=1$	need larger α
Edge continuity	high	ringing
Time complexity	$O(MNk^2)$	$O(MN\log MN)$
Boundary	Manual padding	FFT Auto

3. Discussion(1) **8-neighbor v.s. 4-neighbor**

The above experiments showed that the 8-neighbor mask can simultaneously detect edges in the vertical, horizontal, and **diagonal** directions, resulting in **richer details** in the sharpened image. However, 8-neighbor also **amplify noise** in the image; 4-neighbor produces cleaner edges but may lose diagonal details. Therefore, I ultimately decided to use **8-neighbor** to achieve the best visual effect.

(2) **Alpha in frequency domain**

original image

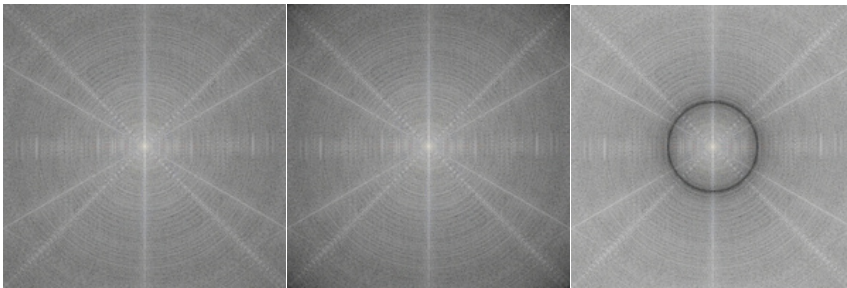
alpha = 1

alpha = 20

I found that when $\alpha=1$, the sharpening result **hardly changes**. I need to increase alpha to around **15~20** to achieve a sharpening effect comparable to that of the spatial domain.

When $\alpha=1$: $H_s=[0,1]$; when $\alpha=20$: by increasing the parameter, the high-frequency components obtain **sufficient gain**. This reflects the importance of considering **numerical scale matching** when designing frequency domain filters, which is also a key difference between spatial and frequency domain implementations.

(3) Fourier Spectrum Black Ring



In the filtered Fourier Spectrum, I observed that when $\alpha = 20$, a distinct **black ring** appeared around the bright spot in the center of the spectrum.

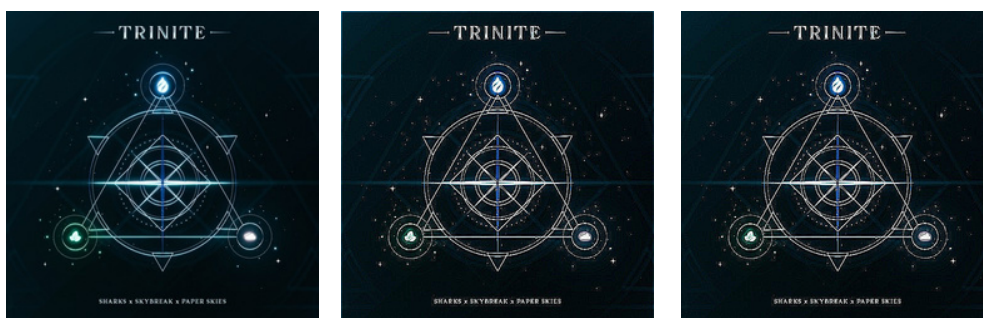
spectrum before $\alpha = 1$ spectrum $\alpha = 20$ spectrum

$$H_{\text{sharpen}}(u, v) = 1 + \alpha \times \left(-\frac{u^2 + v^2}{\max(u^2 + v^2)} \right) \quad \alpha = 20$$

Freq	u^2+v^2	H_sharpen	Potential Impact
Center	0	1.0	Still bright
Mid	~ 0.1	$1 + 20 \times (-0.1) = -1.0$	Negative values cause phase reversal $\rightarrow$ black ring
High	~ 0.8	$1 + 20 \times (-0.8) = -15$	Negative but large value $\rightarrow$ Intensity rebound

I suspect the appearance of a **black ring** might be related to **negative filter values** in the **mid-freq** region, causing **phase reversal** in the frequency components of that region [since uses $\log(1 + |G|)$]. This phenomenon may indicate a potential problem in frequency domain filter design, while an **excessively large** α can effectively enhance high-frequency edges, it may also produce **unexpected phase reversals** in the mid-freq region.

(4) Boundary processing



original image

zero padding

replicate padding

Observing the comparison chart above, the difference between the two padding methods is not significant. Therefore, the report focuses on **zero-padding** as the main result (which is relatively simple to implement).